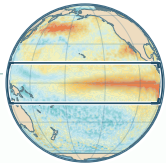
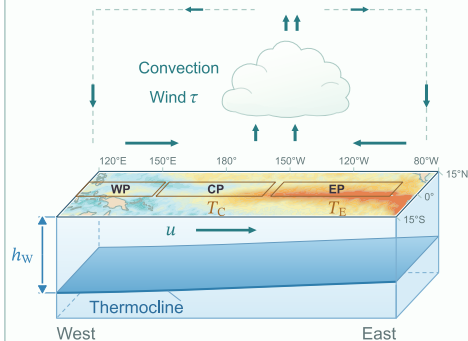


(a) From observations to agent-built models

Observed SST anomalies



Tropical Pacific schematic



Agent-built model

$$d\mathbf{X}_t = \mathbf{f}(\mathbf{X}_t, t) dt + \Sigma(\mathbf{X}_t, t) d\mathbf{W}_t$$

$\mathbf{f}$ Deterministic dynamics Σ Stochastic forcing

$$\mathbf{X}_t = [T_C(t), T_E(t), h_W(t), u(t), \tau(t)]^\top$$

T_C, T_E SST

h_W Thermocline depth

u Ocean current

τ Wind

Task & background
Build an ENSO model.

Observations



SST



Ocean



Wind



AI agent

(b) Final scores of agent-built models

Scientific evaluation

Published reference
0.378

